

Chenxi Ji

Basic Information

Email Address: chenxij2@illinois.edu

Github Account: [Chenxi-Ji](#)

Personal Webpage: <https://chenxi-ji.github.io/>

Education

University of Illinois, Urbana-Champaign, USA

PhD in Computer Engineering

GPA: 4.0/4.0

Sep 2022 - May 2027

Advisor: Mitra Sayan

Tsinghua University, China

Bachelor of Engineering in Automation

GPA: 3.85/4.0

Minor in Data Science

Sep 2018 - Jun 2022

Thesis Advisor: Xiangyang Ji

Research Field

Formal Verification, 3D Reconstruction, Controller Synthesis, Model Compression

Publication

- [Abstract Rendering: Certifying Rendering under 3D Semantic Uncertainty](#). Chenxi Ji*, Yangge Li*, Xiangru Zhong*, Huan Zhang, Mitra Sayan. Accepted as a **Spotlight** paper at *Conf. on Neural Information Processing Systems (NeurIPS)*, 2025.
- [Reachability for Nonsmooth Systems with Lexicographic Jacobian](#). Chenxi Ji, Huan Zhang, Mitra Sayan. Accepted by the proceedings of *Int. Conf. on Tools and Algorithms for Construction and Analysis of Systems (TACAS)*, 2025.
- [Lyapunov Perception Contracts for Operating Design Domains](#). Yangge Li, Chenxi Ji, Jai Anchalia, Mitra Sayan. Accepted by the proceeding of *Learning for Dynamics & Control Conference (L4DC)*, 2025.
- [Verifying Vision-based Autonomy with Abstract Rendering and Perception Contracts](#). Yangge Li, Chenxi Ji, Mitra Sayan. Accepted by the proceedings of *Int. Conf. on Cyber-Physical Systems (ICCPs)*, 2025.
- **Perception Contracts for Reasoning about Reliability of Visual Autonomy: Incremental Stability Certificates**. Chenxi Ji, Yangge Li, Jai Anchalia, Joshua Neighbor, Prashin S. Sharma, Sayan Mitra. Submitted to *ACM Trans. Cyber-Physical Systems (TCPS)*, 2026.

Research Experience

Coordinated Science Laboratory, UIUC

Aug 2022 - Present

Graduate Research Assistant

Advised by Prof. Mitra Sayan

Project: Abstract Rendering --- Certifying Learning-based Rendering Algorithms (Gaussian Splatting and NerF) under Uncertainty from 3D Camera Pose or Scene

Webpage: <https://illinoisreliableautonomygroup.github.io/Abstract-Rendering-Webpage/>

Github: <https://github.com/IllinoisReliableAutonomyGroup/Abstract-Rendering>

Project: Incremental Stability Theorem on Vision-based Formation Flying Tasks

Project: Lyapunov-based Invariance and Stability for Set-Valued Systems

Project: Reachability Analysis of NonSmooth Dynamical Systems via CROWN

Project: Invariance Theorem for Dynamical Systems with Periodic Update Control Signal

Automation Department, Tsinghua University

Oct 2021 - Jun 2022

Undergrad Research Assistant

Advised by Prof. Xiangyang Ji

Project: Lidar-Vision-IMU combined Simultaneous Localization and Mapping (SLAM) and its Application on Autonomous Car

AirLAB, Robotics Institute, Carnegie Mellon University

May 2021 - Dec 2021

Undergrad Research Assistant

Advised by Prof. Sebastian Scherer

Project: Image Sparsity Technology for Rapid 3D Reconstruction using ORB-SLAM2

Dynamic Robotics and Control Lab, USC

Jun 2021 - Sep 2021

Undergrad Research Assistant

Advised by Prof. Quan T Nguyen

Project: Research on Human Robot Dog Interaction Technology Based on Vision

Automation Department, Tsinghua University

Oct 2020 - Jun 2021

Undergrad Research Assistant

Advised by Senior Engineer Geng Lu

Project: Unmanned Aerial Vehicle(UAV) System Design for International Aerial Robotics Competition(IARC)

Institute of Control and Decision Making, Tsinghua University

Sept 2020 – June 2021

Undergrad Research Assistant

Advised by Prof. Xiao He

Project: Simulation and Verification Platform Based on the Three-tank System

Institute of Information Processing, Tsinghua University

Mar 2020 – Sept 2020

Undergrad Research Assistant

Advised by Prof. Feifei Gao

Project: Research on Beam Selection Based on Deep Learning

Teaching

2025 ECE 584 Verification of Embedded Systems, Head Teaching Assistant.

Courses

- Graduate: Control Theory and Design, Nonlinear Control, Interplay Between Control & ML, Embedded System Verification, Trustworthy AI Systems, Pattern Recognition, Radom Process, Adv Topics in Machine Learning
- Undergrad-CS: Data Structure, Computer Language and Programming, Computer Network and Applications, Computer Principles and Applications, Digital Image Processing, Signal and System Analysis, Foundation of Artificial Intelligence, Theory of Automatic Control
- Undergrad-Math: Calculus A, Linear Algebra, Adv Topics in Linear Algebra, Probability and Statistics, Introduction to Complex Analysis, Numerical Analysis and Algorithm

Skills

Programming Skills: 5+years in Python, C/C++,

Software and Tools: PyTorch, Tensorflow, Matlab, CARLA, CROWN, CORA, JuliaReach